ELIZA · embedding experiment on the val split (KAN-97, KAN-221, KAN-99)

Val questions scored: 27

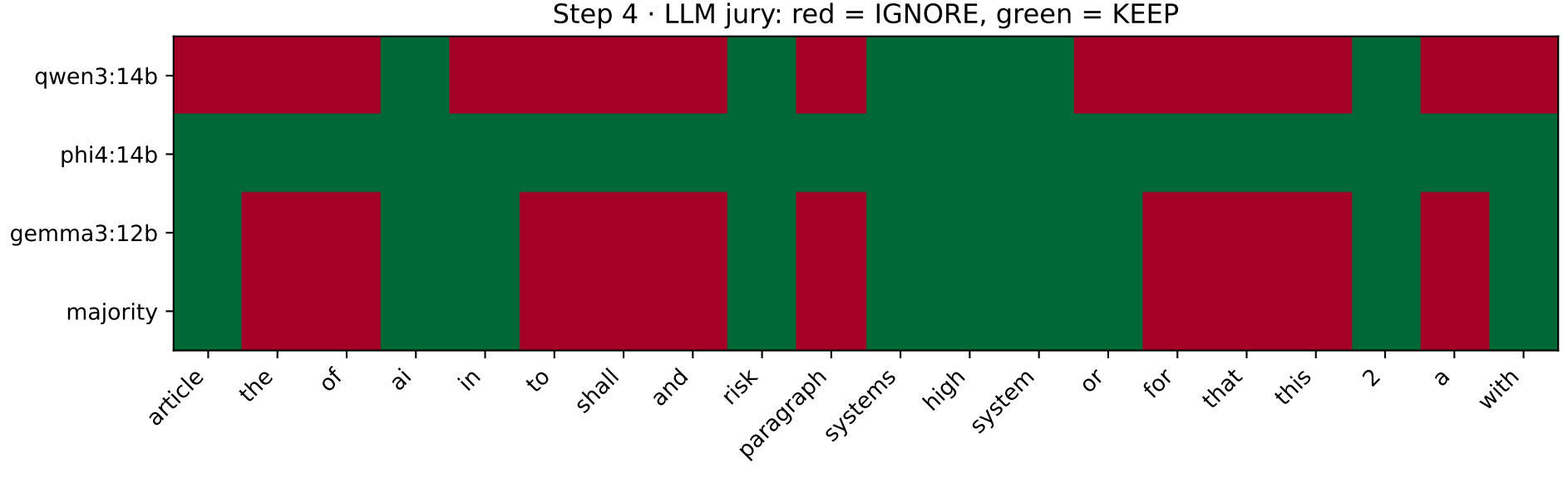
Embedding model: BAAI/bge-m3 dense weight: 0.8 BM25 words removed: stopwords + jury

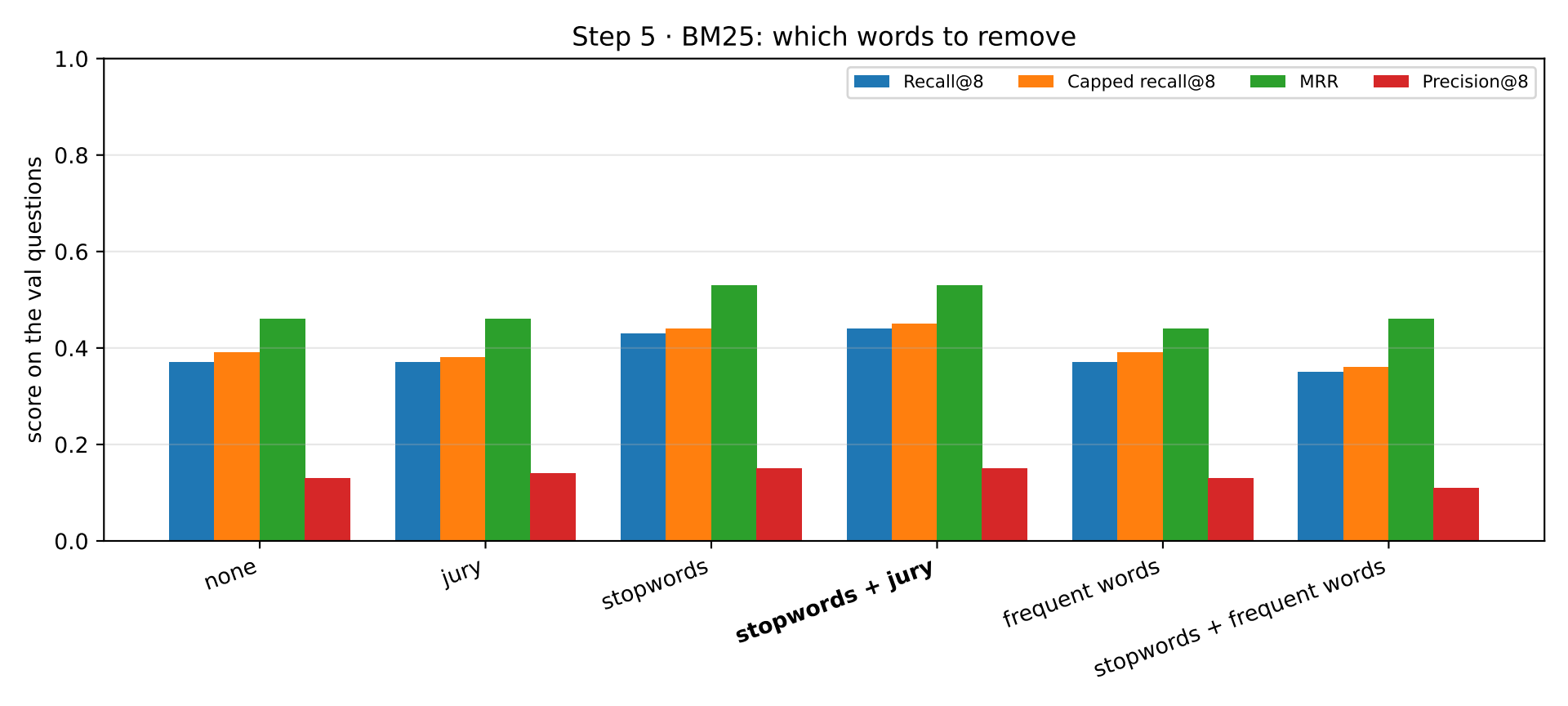
Words removed before embedding: none top-k: 8

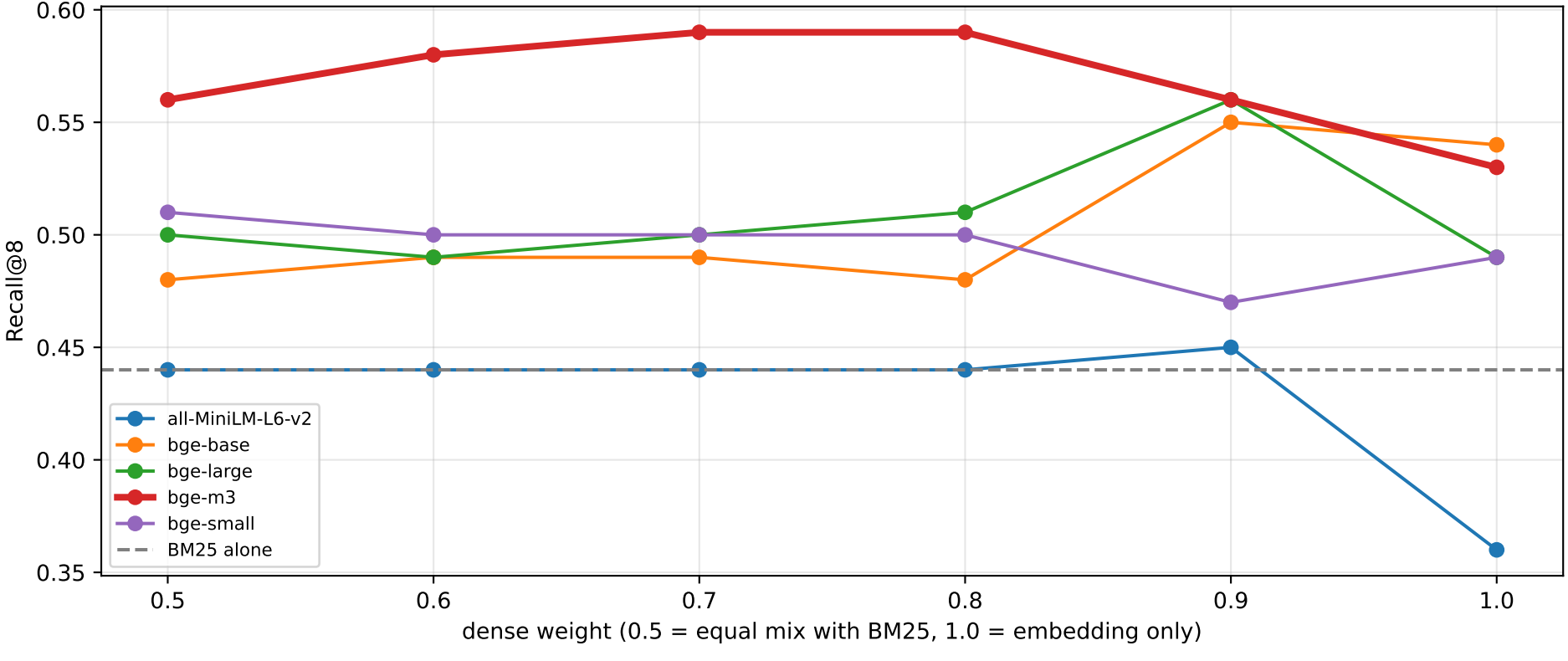
Final search, top 5: Recall@5 0.43, Capped recall@5 0.47, MRR 0.6, Precision@5 0.24

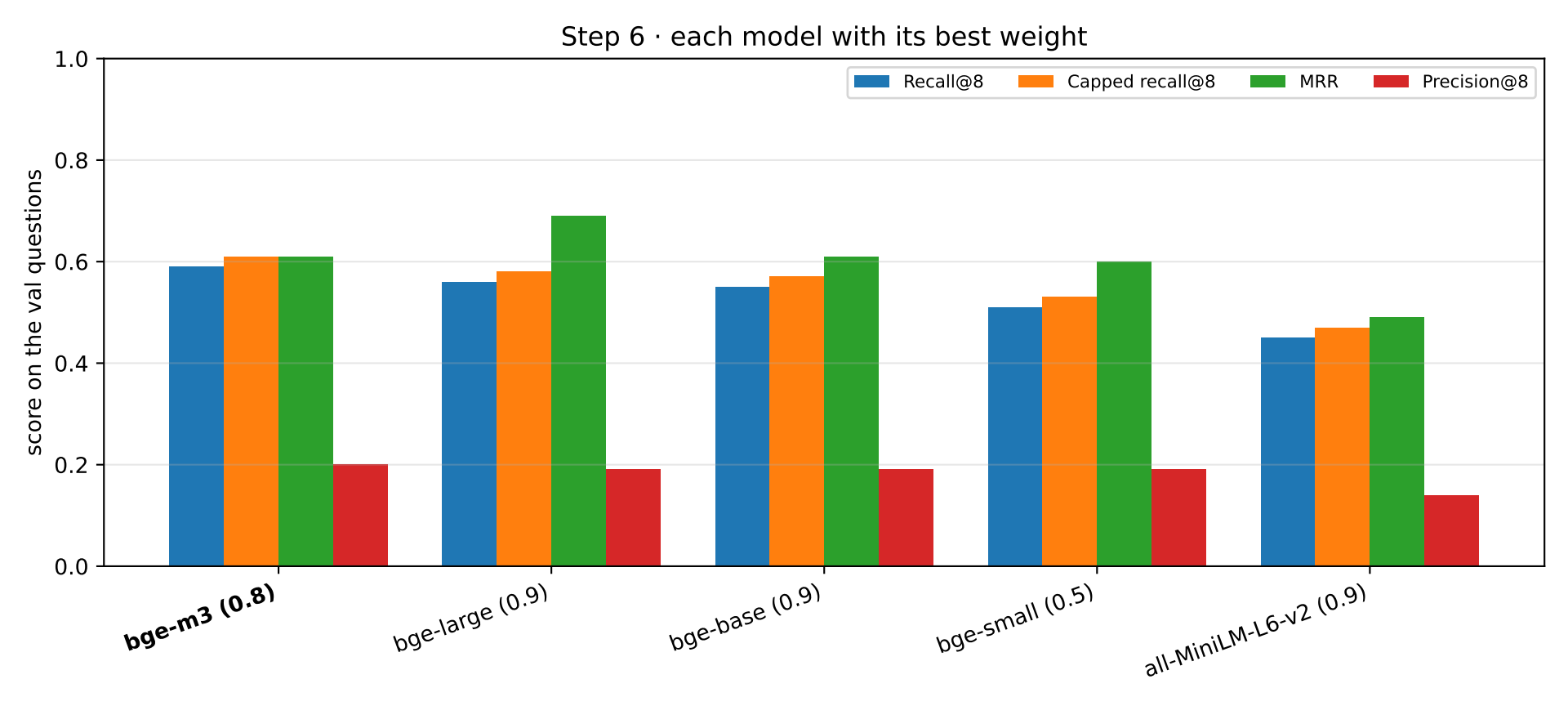
Final search, top 8: Recall@8 0.59, Capped recall@8 0.61, MRR 0.61, Precision@8 0.2

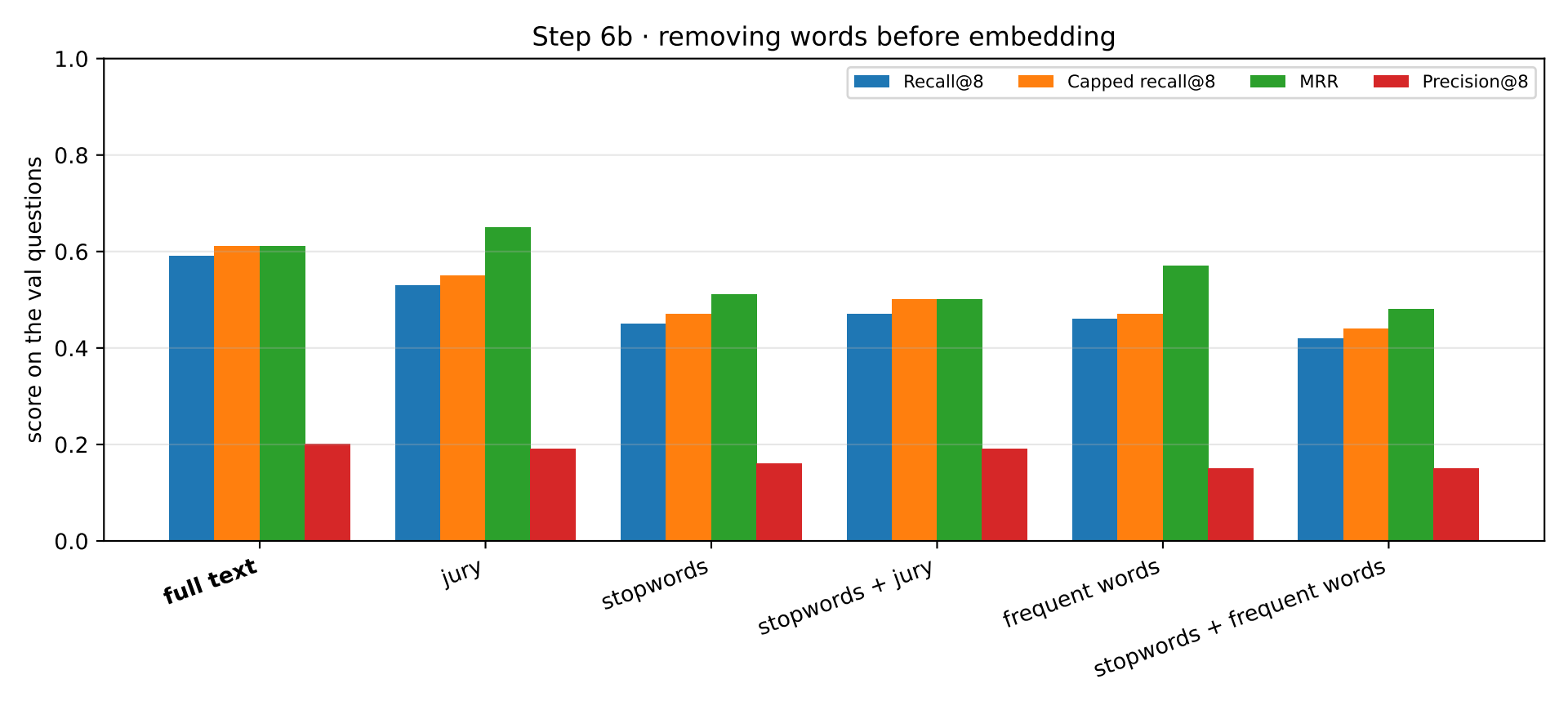
Test split SHA-256: f68bfa569b69783b… commit: fbe9bf8

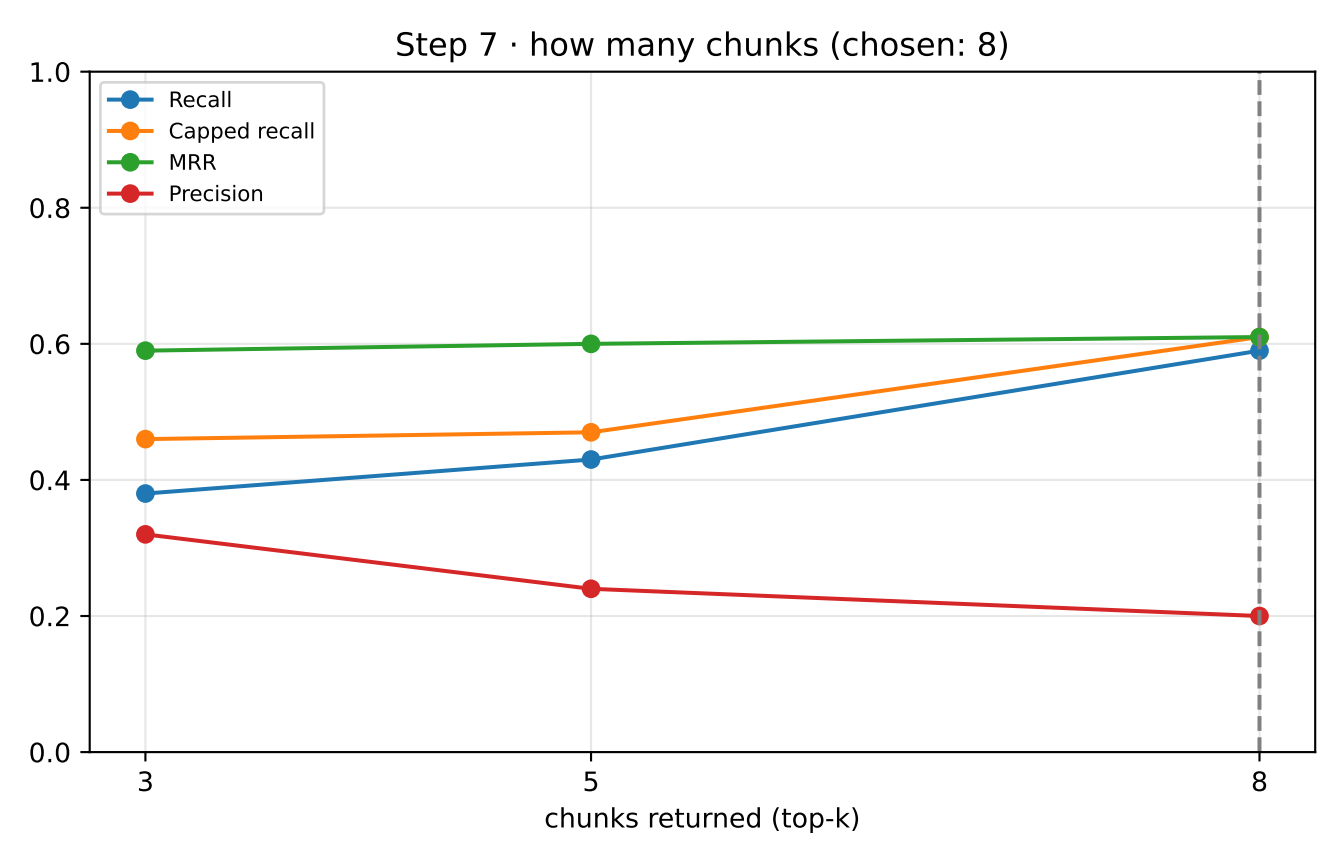

Step 4 · LLM jury: red = IGNORE, green = KEEP
qwen3:14b
phi4:14b
gemma3:12b
majority
article
the
of
ai
in
to
shall
and
risk
paragraph
systems
high
system
or
for
that
this
2
a
with

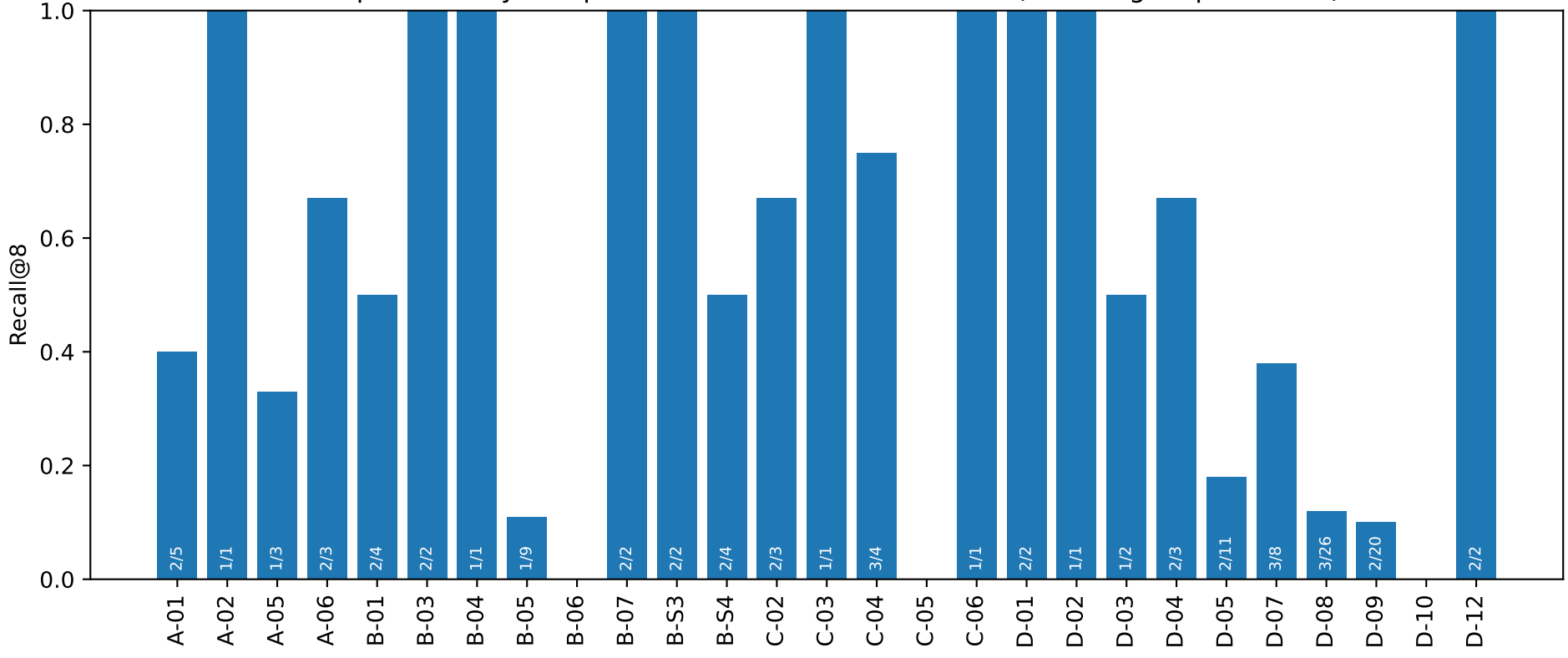

Step 5 · BM25: which words to remove
Recall@8
Capped recall@8
MRR
Precision@8
score on the val questions
1.0
0.8
0.6
0.4
0.2
0.0
none
jury
stopwords
stopwords + jury
frequent words
stopwords + frequent words

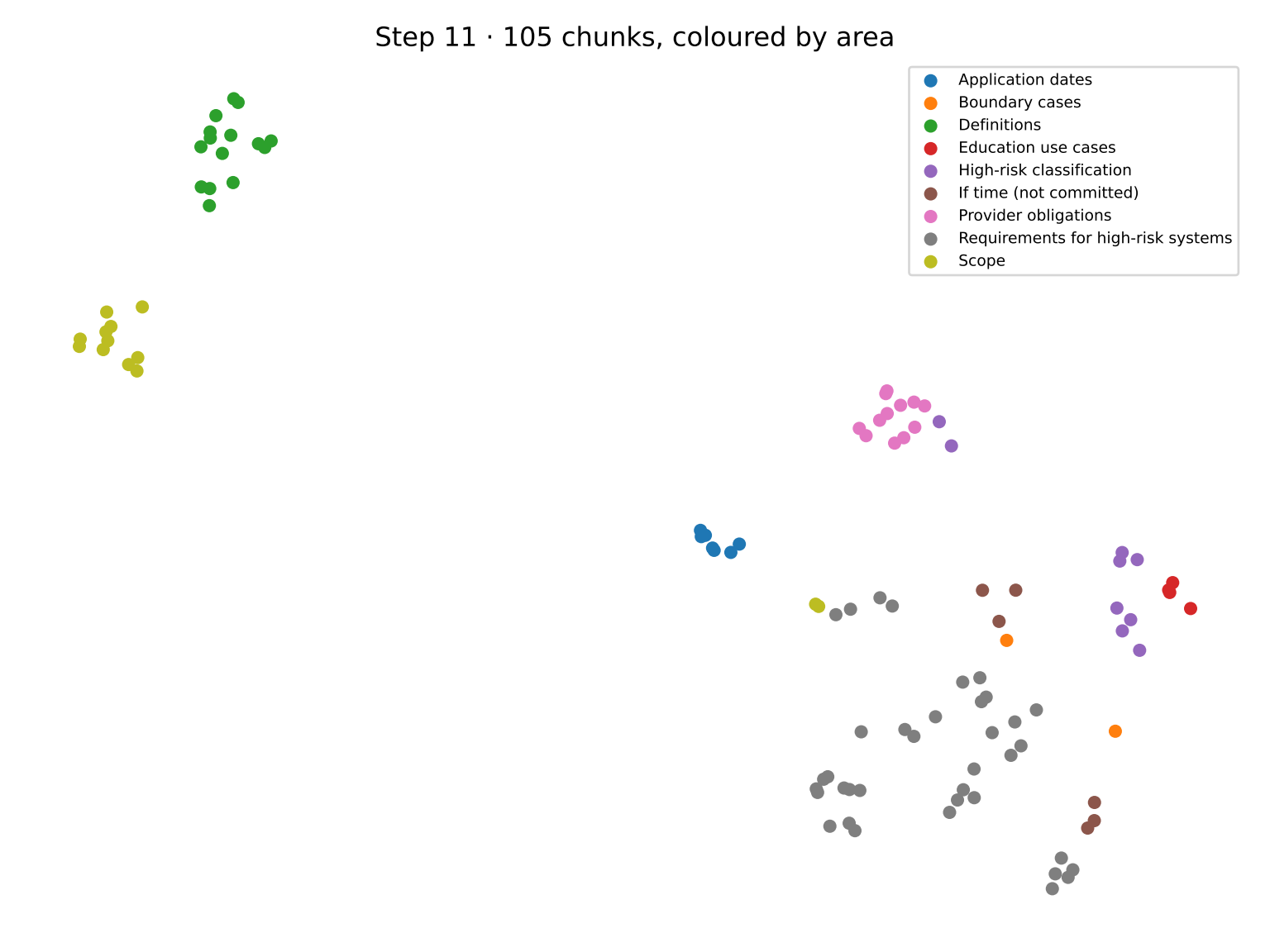

Step 6 · embedding models and BM25 weight (chosen: bge-m3, 0.8)
Recall@8
0.60
0.55
0.50
0.45
0.40
0.35
0.5
0.6
0.7
0.8
0.9
1.0
dense weight (0.5 = equal mix with BM25, 1.0 = embedding only)
all-MiniLM-L6-v2
bge-base
bge-large
bge-m3
bge-small
BM25 alone


Step 6 · each model with its best weight
Recall@8
Capped recall@8
MRR
Precision@8
score on the val questions
1.0
0.8
0.6
0.4
0.2
0.0
bge-m3 (0.8)
bge-large (0.9)
bge-base (0.9)
bge-small (0.5)
all-MiniLM-L6-v2 (0.9)


Step 6b · removing words before embedding
Recall@8
Capped recall@8
MRR
Precision@8
score on the val questions
1.0
0.8
0.6
0.4
0.2
0.0
full text
jury
stopwords
stopwords + jury
frequent words
stopwords + frequent words


Step 7 · how many chunks (chosen: 8)
Recall
Capped recall
MRR
Precision
1.0
0.8
0.6
0.4
0.2
0.0
3
5
8
chunks returned (top-k)


Step 10 · every val question with the chosen search (found / gold provisions)
Recall@8
1.0
0.8
0.6
0.4
0.2
0.0
2/5
1/1
1/3
2/3
2/4
2/2
1/1
1/9
2/2
2/2
2/4
2/3
1/1
3/4
1/1
2/2
1/1
1/2
2/3
2/11
3/8
3/26
2/20
2/2
A-01
A-02
A-05
A-06
B-01
B-03
B-04
B-05
B-06
B-07
B-S3
B-S4
C-02
C-03
C-04
C-05
C-06
D-01
D-02
D-03
D-04
D-05
D-07
D-08
D-09
D-10
D-12


Step 11 · 105 chunks, coloured by area
Application dates
Boundary cases
Definitions
Education use cases
High-risk classification
If time (not committed)
Provider obligations
Requirements for high-risk systems
Scope